

PLATFORM · ENGINEERING

Stack architecture and program flow

One Next.js deployable unit powers the guest marketplace, host console, and ops tools — with MongoDB at the center and a small set of focused external services.

Next.js · App Router

React 19

MongoDB Atlas

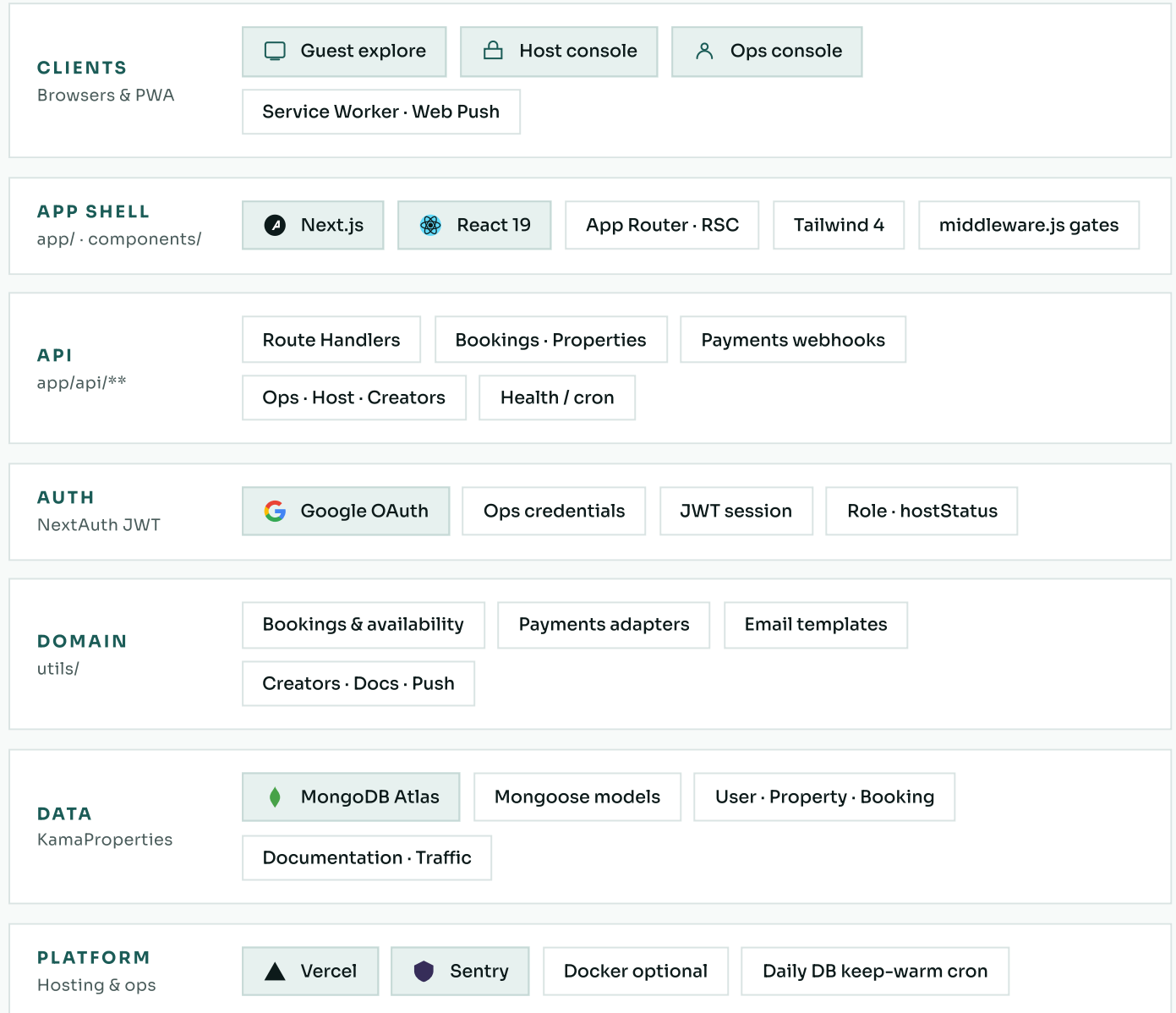
NextAuth

Vercel

PWA

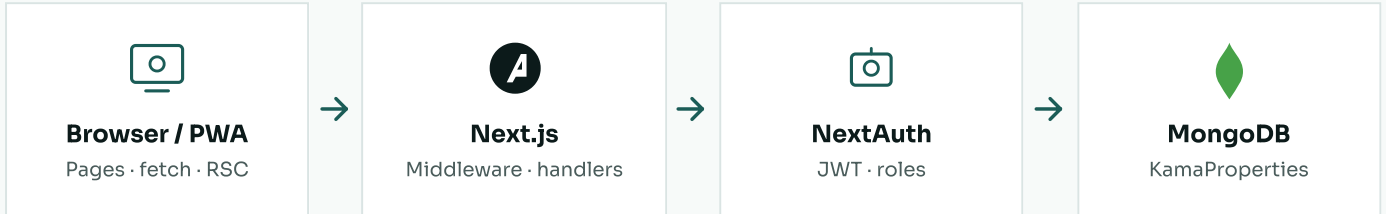
Monolith layers

No separate backend service. Pages, Route Handlers, Mongoose models, and domain utils ship as one Vercel (or Docker) unit.



How a request moves

Typical path for marketplace and console traffic — from the browser through auth and domain logic into MongoDB, with optional side effects.



Three programs, one stack

Guest marketplace, host workspace, and ops console share models and APIs — chrome and auth gates differ.

 Guest / · /properties	 Host /host/*	 Ops /ops/* · /documentation
1. Explore home map & search via GET /api/properties. 2. Open listing /properties/[id] with Maps + media. 3. Request stay POST /api/bookings/request (phone, manual pay). 4. Confirm offline WhatsApp / host — emails & push optional. 5. Save bookmarks on /saved-properties.	1. Onboard /host/onboarding → pending until verified. 2. Home rings arrivals, departures, in-stay, needs-you. 3. Calendar drag/resize availability blocks. 4. Listings create/edit; ops moderates publish. 5. Creators partnership tools under /host/creators.	1. Login credentials provider at /ops/login. 2. Moderate users, listings, reservations, transactions. 3. Market creators, investors, acquisition copilot. 4. Analytics PDF/CSV reports from ops analytics utils. 5. Docs KB seeded Mongo articles on /documentation.
Gate: Google sign-in for booking identity; catalog may be beta-closed.	Gate: hostStatus == "verified" (except install / onboarding).	Gate: role admin / superadmin; docs optional email restrict.

Shell chrome: guest floating tab bar · host top nav · ops side rail + sticky mobile bottom nav. Shared database and domain modules keep the three programs aligned across deploys.

External services

Side-effect services sit beside the monolith. Domain code in `utils/` owns adapters so Route Handlers stay thin.

 Cloudinary Listing and avatar media upload, transforms, and folder provisioning under <code>utils/cloudinary</code> .	 Google Maps Map pins on home/search and Places autocomplete for address entry via <code>utils/googleMaps.js</code> .
 Resend Transactional and marketing email from verified @isisel.com templates (bookings, moderation).	 Payments GeniusPay preferred for ops MoMo; Creem demoted. Guests today: WhatsApp + manual confirmation.
 Sentry Error and performance monitoring across Node, edge, and browser for the Next.js app.	 Vercel Production hosting, Analytics, Blob storage, and scheduled cron for Atlas keep-warm.

SOURCE OF TRUTH Ops docs KB: application-, database-, and api-architecture under <code>/documentation</code> .	THIS DECK docs/isisel-stack-architecture.html → PDF via Chrome print script.	BRAND Teal Ankh mark · accent #1B5C57 · ink #0C1A1A.
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